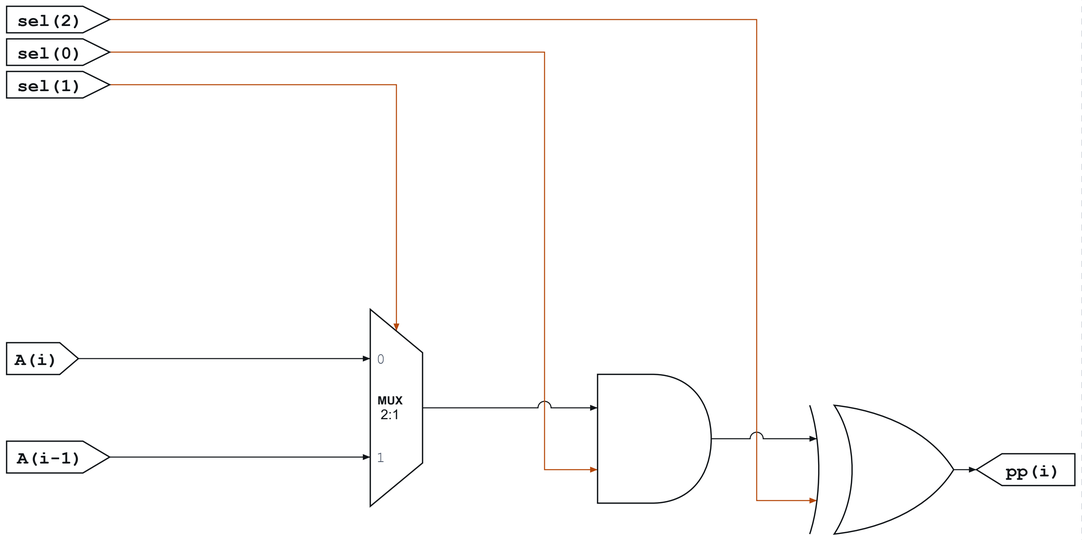
no_sign_extend (steps 2 and 3)

select A or 2A, force zero, then negate with an XOR



per bit: 1 mux + 1 AND + 1 XOR, for each of the N+1 bits of the row
sel(2) drives N+1 XOR gates: a separate negation layer
bit N (the sign) is inverted after the XOR; bits above N are 0
mux input 0 = A(i) (1 x A), input 1 = A(i-1) (2 x A, shifted)

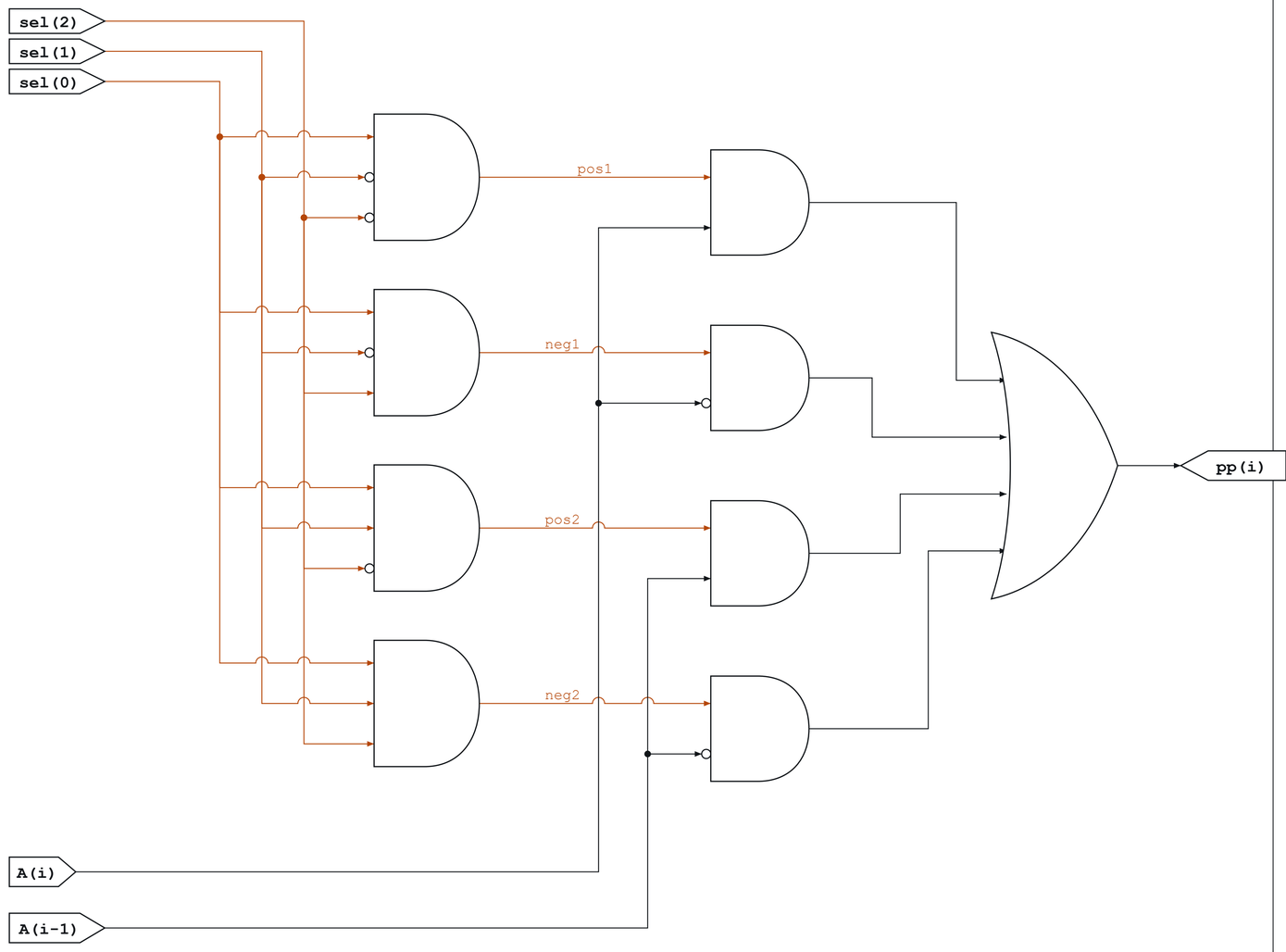
booth_encoder output

digit	sel(2)	sel(1)	sel(0)
0	0	0	0
+1	0	0	1
+2	0	1	1
−1	1	0	1
−2	1	1	1

sel(0) = 1: the digit is not 0
sel(1) = 1: |digit| = 2
sel(2) = 1: the digit is negative

fused_selector (step 4)

decode the four non-zero digits once, then one AND-OR per bit



decoder (left): once per row, shared by the N+1 bits of the row
slice (right): per bit 4 AND2 + 1 OR4, which maps onto AOI/OAI cells
no XOR: the negation is the inverted data input of neg1 / neg2
src1(i) = A(i), src2(i) = A(i-1); bit N inverted as on the left
pos1 = sel(0) . /sel(1) . /sel(2) (+A)
neg1 = sel(0) . /sel(1) . sel(2) (-A)
pos2 = sel(0) . sel(1) . /sel(2) (+2A)
neg2 = sel(0) . sel(1) . sel(2) (-2A)